AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)
Faculty of Engineering
Department of Electrical and Electronic Engineering
Undergraduate Program

PART A

1.	Course No/Course Code	EEE 2103
2.	Course Title	Electronic Devices
3.	Course Type	Core Course
4.	Year/Level/Semester/Term	Second year (4th Semester)
5.	Academic Session	Summer 2024-25
6.	Course Teachers/Instructors	Dr. Wahida Chowdhury, Dr. Mohammad Alif Arman, Dr. Md. Rifat Hazari, Ms. Sadia Yasmin, Dr. Nadia Afroze, Ms. Samiha Ishrat Islam, Dr. Ebad Zahir, Prof. Dr. Md. Nasir Uddin, Ms. Lia Moni, Dr. M. Tanseer Ali, Mr. Md. Mahmudul Hasan, Mr. Joheb Muhammad Tanzeer Sayeed, Ms. Tahseen Asma Meem
7.	Pre-requisite (If any)	EEE 2101: Electrical Circuits 2 (AC)
8.	Credit Value	3 credit hours
9.	Contact Hours	4 hours of theory per week
10.	Total Marks	100
	Mission of EEE Department	Educate young leaders for academia, industry, entrepreneurship, and public and private organization through theory and practical knowledge to solve engineering problems individually and in teams. Create knowledge through innovative research and collaboration with multiple disciplines and societies. Serve the communities at national, regional, and global levels with ethical and professional responsibilities.
11.	Vision of EEE Department	To become a front runner in preparing Electrical and Electronics Engineering graduates to be nationally and globally competitive and thereby contribute value for the knowledge-based economy and welfare for the people of the world.
12.	Rationale of the Course (Course Description)	The Electronic Devices course provides foundational knowledge on semiconductor components like diodes, BJTs, and FETs, essential for understanding modern electronic circuits. It supports the analysis and design of systems used in communication, computing, and power electronics. This course is vital for progressing in advanced subjects such as analog electronics, VLSI, and microelectronics.
13.	Course Content	The course is designed to provide students' knowledge with: • **Semiconductor Basics:** Intrinsic and extrinsic materials, energy bands, p-n junctions, diode characteristics, Zener diodes, and equivalent circuits. • **Diode Applications:** Load-line analysis, series-parallel diodes, rectifiers (half-wave, full-wave), clippers, clampers, logic gates using diodes, and voltage regulation. • **BJT Devices:** CB and CE configurations, DC biasing methods, small-signal models, and low-frequency AC analysis. • **FET Devices:** JFET, D-MOSFET, and E-MOSFET construction, biasing techniques, and amplifier configurations.

15. Course Outcomes (CO)/Course Learning Outcomes (CLOs):

By the end of this course, students should be able to –

COs/ CLOs Number	COs/CLOs Statements	K	P	A	Assessed Program Outcome Indicator	BNQF Indicator	Teaching - Learning Strategy	Assessment Strategy
1	**Identify the semiconductor diode principles with comparison to each other for practical applications having different electronic arrangements.**	**1**			**P.b.1.C4**	**FS.1**	**Lecture Tutorial**	**OBE Assignment (Mid)**
2	Apply information and concepts of mathematics to solve basic electronic circuits e.g., series, parallel, clipper, clamper, and Zener diode circuits and DC biasing of BJT circuits.	2			P.a.2.C3	FS.2	Lecture Tutorial	Quiz and Term Exam (Mid)
3	Formulate solutions, procedures, and methods of BJT circuits using AC analysis concepts.	2			P.b.2.C4	FS.1	Lecture Tutorial	Term Exam (Final)
4	Formulate solutions, procedures, and methods of FET circuits using DC and AC analysis concepts	2			P.b.2.C4	FS.1	Lecture Tutorial	Quiz and Term Exam (Final)

16. Mapping with Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
1		FS.1										
2	FS.2											
3		FS.1										
4		FS.1										

PART B

17. Course plan:

By the end of this course, students should be able to –

Time Frame (Week)	Topics	Teaching Learning Strategy	Assessment Strategy	Corresponding COs /CLOs	Assessment
Week 1	Mission & Vision of AIUB, Dept. of EEE; Semiconductor materials, Energy levels, Intrinsic and extrinsic materials (n- and p-type), Biasing of p-n junctions,	Lecture Tutorial	Judging the skill of concepts and electronic designs through	CO1	OBE Assignment, Quiz and term Exam (Mid)

	Semiconductor diodes, I-V characteristics, Zener region, Resistance levels. Diode equivalent circuits, Load-line analysis, Diode-Approximations, Series, Parallel and combination of Series-Parallel diode configurations,		fundamental understanding and numerical problems using assignments, quizzes, and mid-term examination.		
Week 2	Half-wave Rectification, Full-wave Rectification: Full bridge rectifier, Center-tap rectifier.	Lecture Tutorial		CO2	
Week 3	Clippers and Clamper circuits. Zener diodes, voltage regulation with fixed and variable voltage/loads (Network Analysis). Bipolar Junction Transistor (BJT): Construction, operation and characteristics of Common-Base (CB) configuration.	Lecture Tutorial		CO2	
Week 4	Bipolar Junction Transistor (BJT): Construction, operation and characteristics of Common-Emitter (CE) configuration. BJT DC biasing: Operating point, Fixed-Bias and Emitter-Bias configuration. BJT DC biasing: Voltage-Divider Bias, Collector feedback, Emitter-Follower configuration.	Lecture Tutorial		CO2	
Week 5	**MID-TERM EXAM WEEK**				
Week 6	BJT small signal low frequency AC response; BJT transistor modeling: r_e transistor model; CB and CE configurations, CE fixed bias, Voltage-Divider bias, Emitter bias, Emitter-Follower, CB amplifier.	Lecture Tutorial	Judging skill of concepts and electronic designs through critical understanding, numerical and analytical problems using quizzes and final examination	CO3	Term Exam (Final)
Week 7	JFET: Construction and characteristics, Depletion and Enhancement Type MOSFET: Construction, operation and characteristics, CMOS inverter.	Lecture Tutorial		CO4	Quiz and Term Exam (Final)
Week 8	JFET DC Biasing: Fixed-Bias and Self-Bias configuration, Voltage-Divider Bias, D-MOSFET DC Biasing: Fixed Bias configuration, Self-Bias configuration, E-MOSFET DC biasing: Feedback biasing and Voltage-Divider biasing configuration.	Lecture Tutorial		CO4	
Week 9	JFET small signal model, JFET input and output impedance, JFET AC equivalent circuit.	Lecture Tutorial		CO4	

	JFET Amplifier: Fixed bias configuration. JFET Amplifier: Self-bias configuration, Voltage divider configuration. JFET Amplifier: Common-Gate configuration, Common-Drain configuration, D-MOSFET amplifiers.				
Week 10	E-MOSFET amplifiers: Drain-Feedback configuration, Voltage-Divider configuration.	Lecture Tutorial		CO4	
Week 11	**FINAL-TERM EXAM WEEK**				

* The faculty reserves the right to change, amend, add or delete any of the contents.

PART C

18. Assessment and Evaluation

1. Assessment Strategy:

	CO/CLO 1 (marks)	CO/CLO 2 (marks)	CO/CLO 3 (marks)	CO/CLO 4 (marks)	Marks for Grading
Assignment (Mid)	20				20
Quiz (Mid)		30			30
Term exam (Mid)		40			40
Quiz (Final)				50	50
Term exam (Final)			10	30	40
Total	**20**	**70**	**10**	**80**	**180**

2. Table of Specification (TOS)

<u>Mid-Term Exam</u>

					Level of Bloom's Taxonomy																		
					Remember			Understand			Apply			Analyze			Evaluate			Create			
Topics	CO No.	No. of Days	No. of Items	No. of COs	Item No.	Test Type	Marks	Item No.	Test Type	Marks	Item No.	Test Type	Marks	Item No.	Test Type	Marks	Item No.	Test Type	Marks	Item No.	Test Type	Marks	POI
Semiconductor Diode Basics	CO1	2	0	`																			P.b.1.C4
Load-Line Analysis	CO2	1	1								1(a)	PS	5										P.a.2.C3
Series-Parallel Diodes	CO2	1	2								2(a)	PS	5										
											5(b)	PS	5										
Rectifiers	CO2	1	1								3(a)	PS	5										
Clippers	CO2	1	1								4(a)	PS	5										
Clampers	CO2	1	1								5(a)	PS	5										
Zener Diode	CO2	1	2								1(b)	PS	5										
											4(b)	PS	5										
BJT DC Biasing	CO2	4	2								2(b)	PS	5										
											3(b)	PS	5										
Total	1	12	10										50										

Final Exam

					Level of Bloom's Taxonomy																		
					Remember			Understand			Apply			Analyze			Evaluate			Create			
Topics	CO No.	No. of Days	No. of Items	No. of COs	Item No.	Test Type	Marks	Item No.	Test Type	Marks	Item No.	Test Type	Marks	Item No.	Test Type	Marks	Item No.	Test Type	Marks	Item No.	Test Type	Marks	POI
BJT AC Biasing	CO3	2	2											1(a)	PS	5							P.b.2.C4
														1(b)	PS	5							
FET Biasing	CO4	6	6											2(a)	PS	5							
														2(b)	PS	5							
														3(a)	PS	5							
														3(b)	PS	5							
														4(a)	PS	5							
														5(a)	PS	5							
FET Amplifiers	CO4	4	2											4(b)	PS	5							
														5(b)	PS	5							
Total		14	10													50							

***Test Type Legend**: **AS:** Assignment; **BQ**: Broad question; **SQ**: Short question; **D**: Derivation; **ES:** Essay; **EX:** Exercise; **GE:** Group Exercise; **ID:** Identification; **MC**: Multiple Choice; **MT**: Matching Type; **OB:** Observation; **PS**: Problem Solving; **SA**: Short Answer; **TF**: True or False; **VV:** Viva Voce; **Other please specify**:*

3. Marks Distribution:

The evaluation system will be strictly followed as par the AIUB grading policy. The following grading system will be strictly followed in this class.

Assessment Type	Marking system For Theory Classes (Midterm)	
Continuous	Attendance	10%
Continuous	OBE Assignment	20%
Continuous	Quiz	30%
Summative	Midterm Exam	40%
	Total	**100%**
Assessment Type	**Marking system For Theory Classes (Final term)**	
Continuous	Attendance	10%
Continuous	Quiz	50%
Summative	Final term Exam	40%
	Total	**100%**
Final Grade/ Grand Total		
Grand Total	Midterm:	40%
	Final Term:	60%

4. **Grading Policy**

Letter	Grade Point	Numerical %
A+	4.00	90-100
A	3.75	85-<90
B+	3.50	80-<85
B	3.25	75-<80
C+	3.00	70-<75
C	2.75	65-<70
D+	2.50	60-<65
D	2.25	50-<60
F	0.00	<50(Failed)

5. **Makeup Procedure:**

Students who fail to maintain the requirements and deadlines needed to contact faculty with reasoning. Continuous assessments will be taken with agreement with the student and faculty. For the make up of Summative assessments, students need to apply for SET – B exam according to the AIUB policy.

PART D

19. Learning Materials

Formal lectures will provide the theoretical base for the subject as well as covering its practical application. A set of lecture notes, tutorial examples, with subsequent discussion and explanation, together with suggested reading will support and direct the students in their own personal study.
Maximum topics will be covered from the textbook. For the rest of the topics, reference books will be followed. Some Class notes will be uploaded on the web. White board will be used for most of the time.
For some cases, multimedia projector will be used for the convenience of the students.
Students must study up to the last lecture before coming to the class and it is suggested that they should go through the relevant chapter before coming to the class. Just being present in the class is not enough- students must participate in classroom discussions.

Few assignments will be given to the students based on that class to test their class performance.

1. **Recommended Readings (Textbook):**

[1] Robert. L. Boylestad & Louis Nashelky, "Electronics Devices and Circuit Theory", 11th edition, Prentice Hall.

2. **Supplementary Readings (Reference Book):**

[1] Muhammad H. Rashid, "Microelectronic Circuits Analysis and Design", 2nd edition, CL Engineering, 2010.
[2] Adel S. Sedra & Kenneth C. Smith, "Microelectronic Circuit", 5th edition, Oxford University press.
[3] Jimmie J. Cathey, "Schaum's Outline of Electronic Devices and Circuits", 2nd Edition.
[4] Richard S. Muller, Theodore I. Kamins & Mansun Chan, "Device Electronics for Integrated Circuits".
[5] John Henderson, "Electronic Devices: Concepts and Applications".
[6] Ali Aminian & Marian Kazimierczuk, "Electronic Devices: A Design Approach".
[7] Ben. G. Streetman & S.K. Banerjee, "Solid State Electronics", 6th edition, Prentice Hall.
[8] Jacob Millman & Christos C. Halkias, "Integrated Electronics", Tata McGrew-Hill edition.
[9] Paul Horowitz & Winfield Hill, "The Art of Electronics", 2nd Edition, Cambridge University Press.

PART E

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<tr><td colspan="3">Verification: EEE 2103: Electronic Devices</td></tr>
<tr><td>Prepared by:

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Prof. Dr. A B M Siddique Hossain
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